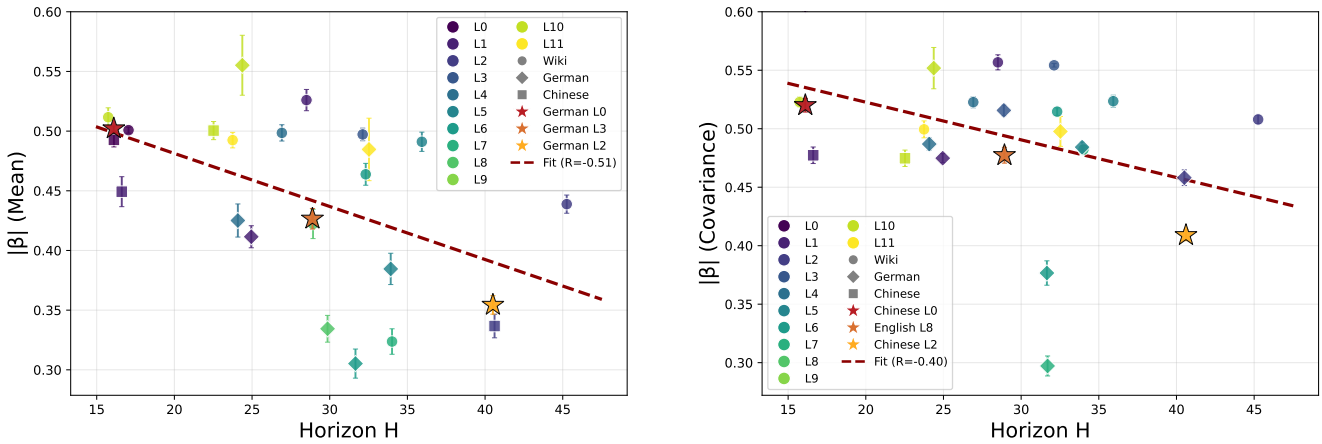
Rebuttal Figures — Additional Experiments

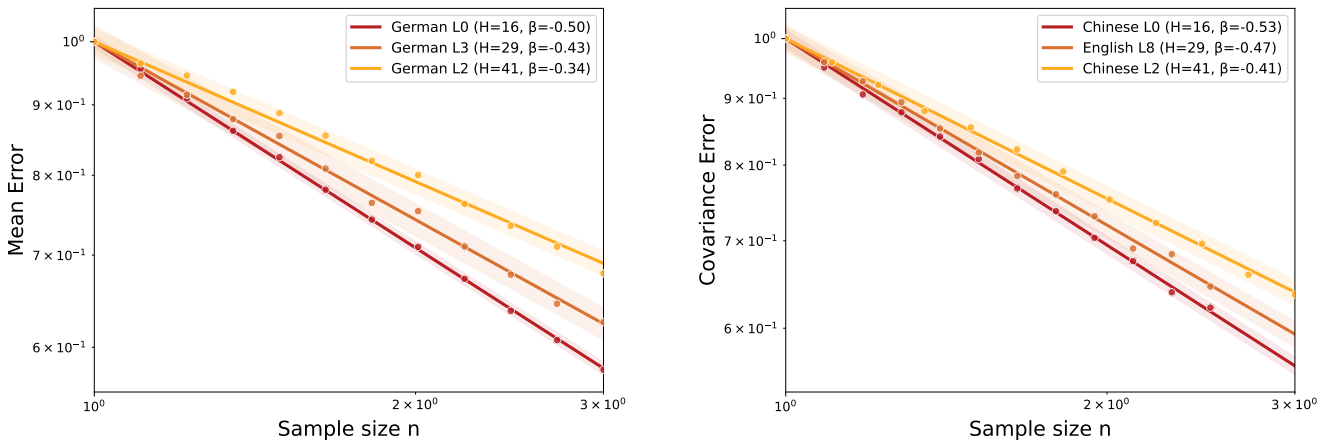
1 BERT experiments (dense attention)



(a) Mean error

(b) Covariance error

Figure 1: **Convergence rate $|\beta|$ versus horizon H across BERT-base-uncased's layers** 0–11 **and all Wikipedia text distributions.** Markers represent (layer, text configuration) pairs; shapes denote text sources (Wikipedia EN/DE/ZH) and colors indicate layers. Larger $|\beta|$ implies faster convergence. The convergence rate $|\beta|$ is estimated via log-log linear regression of the per-token mean (left) or covariance (right) error against n, with $n \in [1000, 5000]$ i.i.d. sampled keys and $k \in [500, 1000]$ Monte Carlo repetitions. Star markers denote the configurations whose detailed convergence curves are displayed in Figure 2. The dashed line is a linear fit across all points.

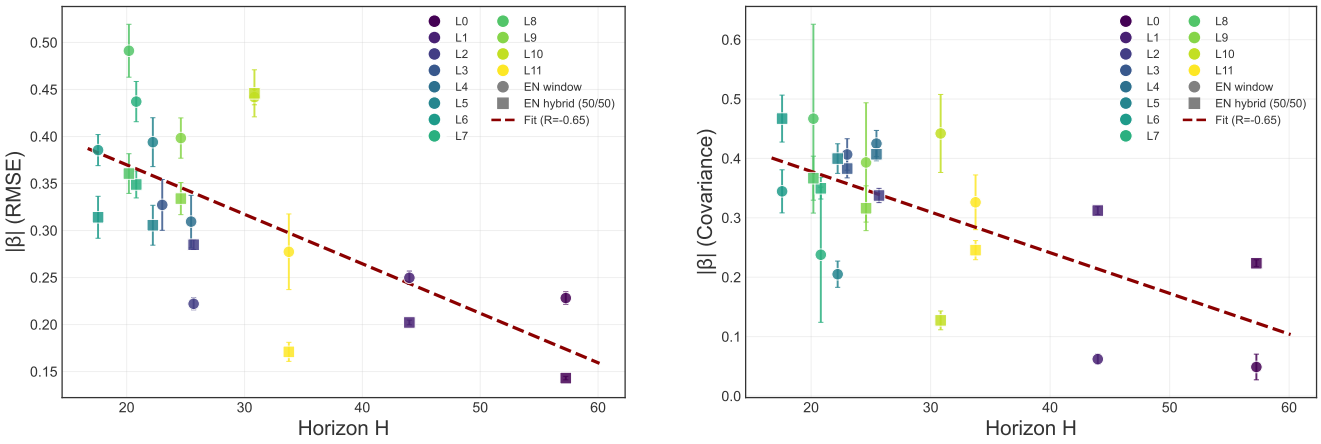


(a) Mean error

(b) Covariance error

Figure 2: **Convergence curves for starred layers.** Mean error (left) and covariance error (right) versus subsample size n for the three starred configurations from Figure 1, using BERT-base-uncased (dense attention). Markers show the averaged ℓ_2 errors over $k = 500$–1000 subsamples, with ± 3 standard-error bands. Log–log linear fits yield power-law exponents $|\beta|$ (legend), with error scaling as $n^{-|\beta|}$. From dark red to light orange as the horizon H increases, confirming the slow-down predicted by Theorem 5.3.

2 Windowed sampling experiment



(a) RMSE error

(b) Covariance error

Figure 3: **Convergence rate $|\beta|$ versus horizon H under non-i.i.d. windowed sampling.** BigBird-RoBERTa's layers 0–11 on English Wikipedia, under two sampling strategies: fixed-random (EN window, $r = 50$ random tokens outside a sliding window of size $n-r-1$) and hybrid (EN hybrid 50/50, half window and half random tokens). Colors indicate layers. The negative trend (fit slope $R = -0.65$) confirms that the slow-down driven by H persists under structured, non-i.i.d. sampling.

3 Synthetic experiment: uniform distribution on the unit sphere

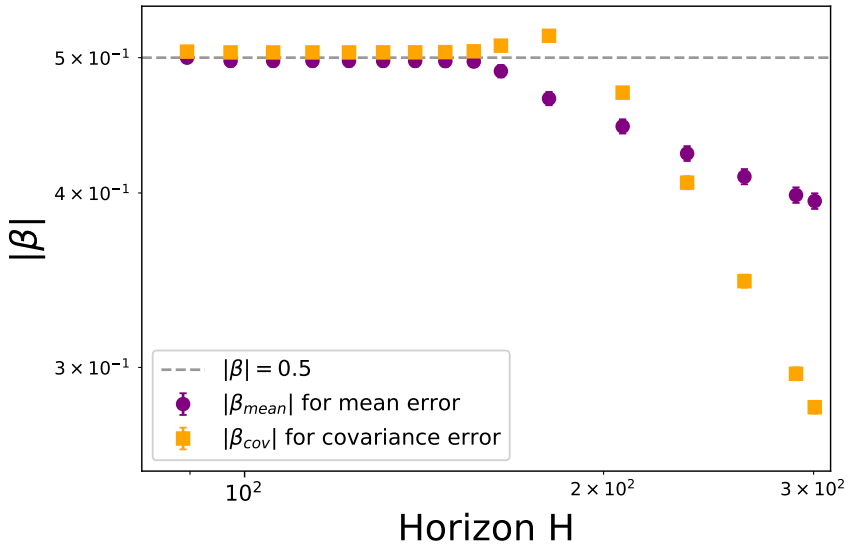


Figure 4: **Convergence rate $|\beta|$ versus horizon H for uniform distributions on the unit sphere.** Empirical convergence rate exponents $|\beta_{\text{mean}}|$ and $|\beta_{\text{cov}}|$ as a function of attention horizon H for token embeddings drawn uniformly from the unit sphere $\mathbb{S}^{d-1}$. The grey dashed line indicates the classical rate exponent $|\beta_0| = 0.5$ corresponding to the parametric rate $1/\sqrt{n}$. The same slow-down as in the Gaussian case is observed as H increases, confirming that the sub-parametric convergence regime predicted by Theorem 5.3 extends beyond Gaussian distributions.

4 Downstream classification experiment

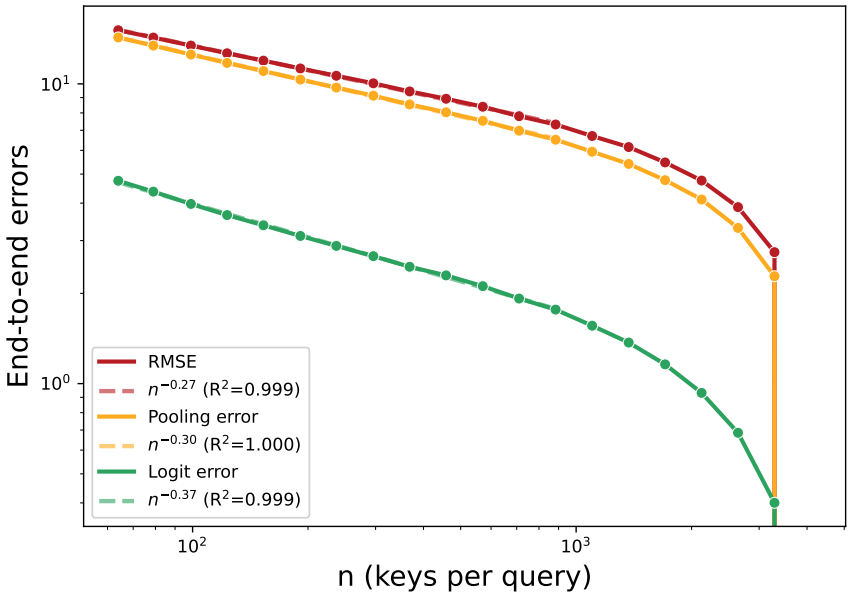


Figure 5: **End-to-end error convergence versus number of keys per query n.** We observe sub-parametric convergence on downstream tasks, with RMSE ($\beta = 0.27$), pooling error ($\beta = 0.30$) and logit error ($\beta = 0.37$) all scaling as $n^{-\beta}$ with $\beta < \frac{1}{2}$, consistent with Theorem 5.3. Experiment: BigBird-RoBERTa-base fine-tuned for 3 epochs with mean pooling on arxiv-classification (11 categories, dense accuracy 86.4%), with n i.i.d. sampled keys per query used to compute attention at each layer. Dashed lines show power-law fits on $n \leq 884$.

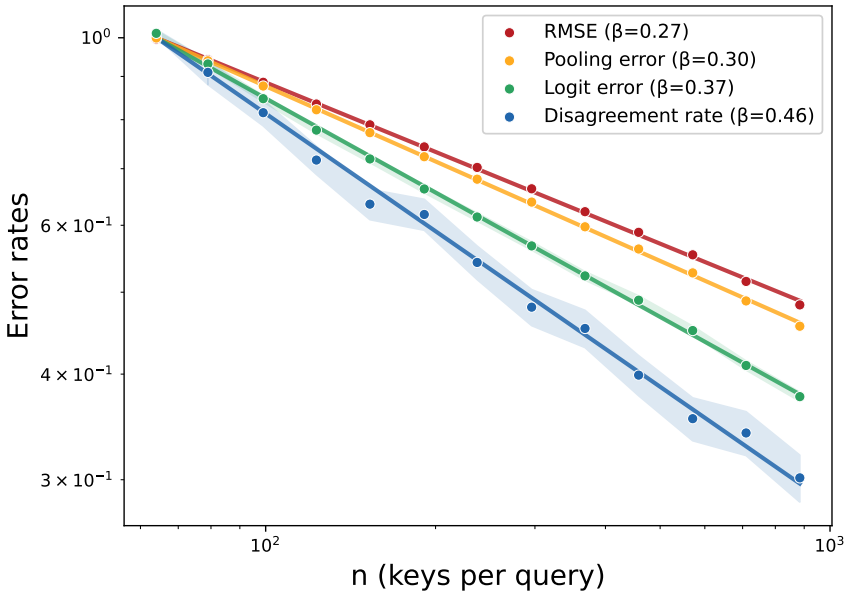


Figure 6: **Normalized error rates versus n.** All downstream error metrics exhibit sub-parametric convergence $n^{-\beta}$ with $\beta < \frac{1}{2}$: RMSE ($\beta = 0.27$), pooling error ($\beta = 0.30$), logit error ($\beta = 0.37$) and disagreement rate — fraction of examples where sparse and dense predictions differ — ($\beta = 0.46$). Downstream metrics converge more rapidly than embedding-level metrics. Shaded bands show ± 3 standard-error intervals.

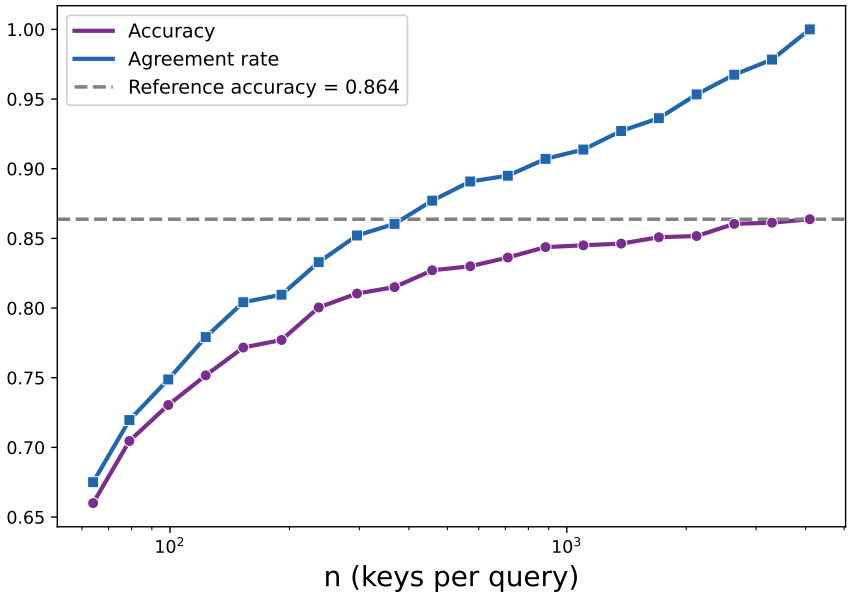


Figure 7: **Classification accuracy and agreement rate versus n.** Accuracy (fraction of correctly classified examples) and agreement rate with the dense model (fraction of examples where the sparse and dense models predict the same label). The reference accuracy (dashed line) is 86.4%. The agreement rate converges to 1 as expected, and the accuracy approaches the dense model's performance with diminishing returns: with only 25% of keys ($n = 1101$ out of 4096), the model already achieves 84.5% accuracy, and doubling the number of keys from $n = 1101$ to $n = 2124$ yields only a +0.7% accuracy gain.

5 Effective rank of the token covariance matrix per layer

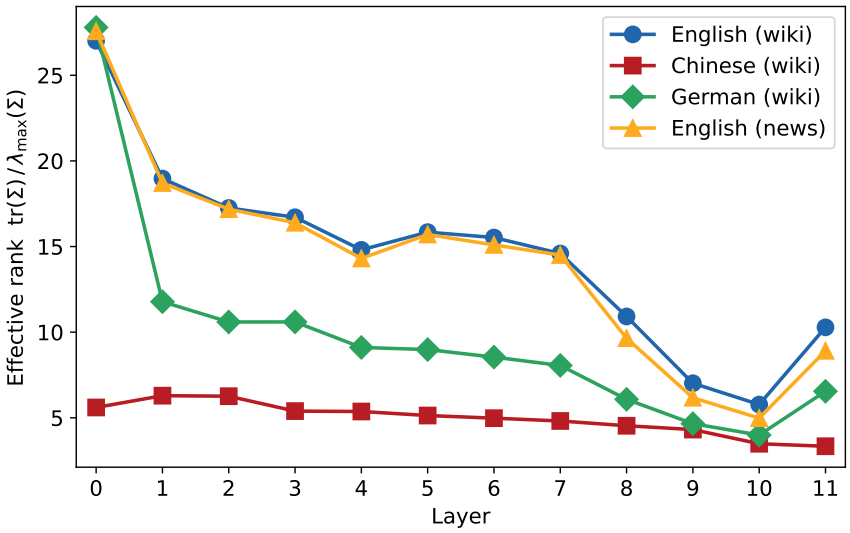


Figure 8: **Effective rank** $\mathrm{tr}(\Sigma)/\lambda_{\max}(\Sigma)$ **of the token covariance matrix at each layer of BigBird-RoBERTa-base, across four datasets.** The effective rank is evaluated on English Wikipedia, Chinese Wikipedia, German Wikipedia, and English news (CC-News), using the same pretrained model weights; differences arise solely from the input distribution. The effective rank decreases with depth across all datasets, indicative of increasing spectral concentration in deeper layers. Chinese Wikipedia exhibits markedly lower effective rank at every layer ($\sim$3–6 vs. $\sim$5–27 for the other datasets), reflecting its higher anisotropy. English Wikipedia and English news overlap closely throughout all layers, suggesting that effective rank might be primarily language-dependent rather than domain-dependent.